SWE202
E-commerce Platform

STATE MACHINE DIAGRAMS & ERD DIAGRAM

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STATE MACHINE DIAGRAMS

The state machine diagrams in this project are designed according to different user levels, illustrating how each type of user interacts with the system and the transitions between various states.

1. Vendor State Machine Diagrams

1.1 Vendor Order Fulfillment

Description

This diagram models how a vendor processes an order from the moment it is assigned until it is completed. It captures the vendor's operational workflow and decision-making during order handling.

Process Flow

The process begins in the Idle state. When a new order is assigned, the vendor transitions to *Order Received*, and then to *Reviewing Order*, where the order is evaluated.

At this stage, a decision is made:

- If the order is approved, the process continues
- If the order is rejected, the process terminates and returns to Idle

If accepted, the vendor proceeds through the following stages:

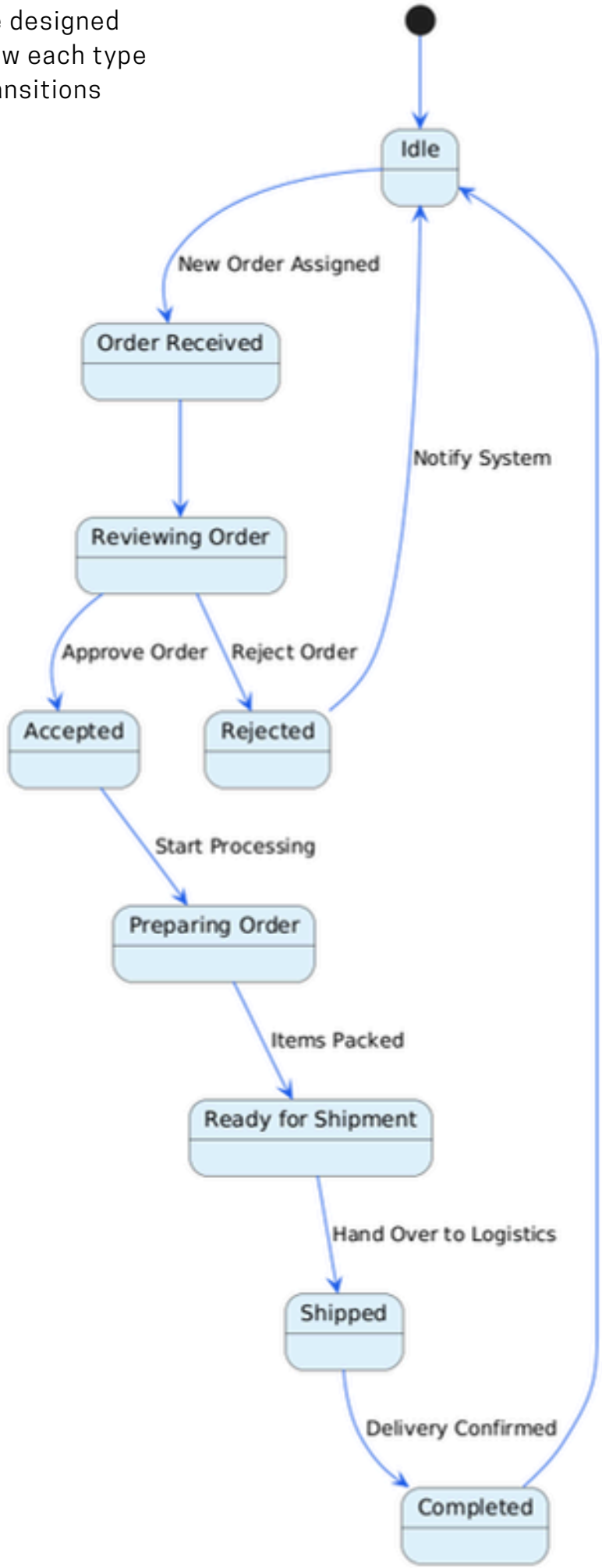
- Preparing Order – items are collected and packed
- Ready for Shipment – preparation is complete
- Shipped – order is handed to the logistics provider
- Completed – delivery confirmation is received

Key Observations

- Includes a decision point (accept/reject), introducing branching
- Logistics provider interaction is reflected in the Shipped state
- Lifecycle ends only after
- delivery confirmation

Design Relevance

- The definition of order status values in the database
- Clear representation of vendor responsibilities
- Integration with external delivery systems



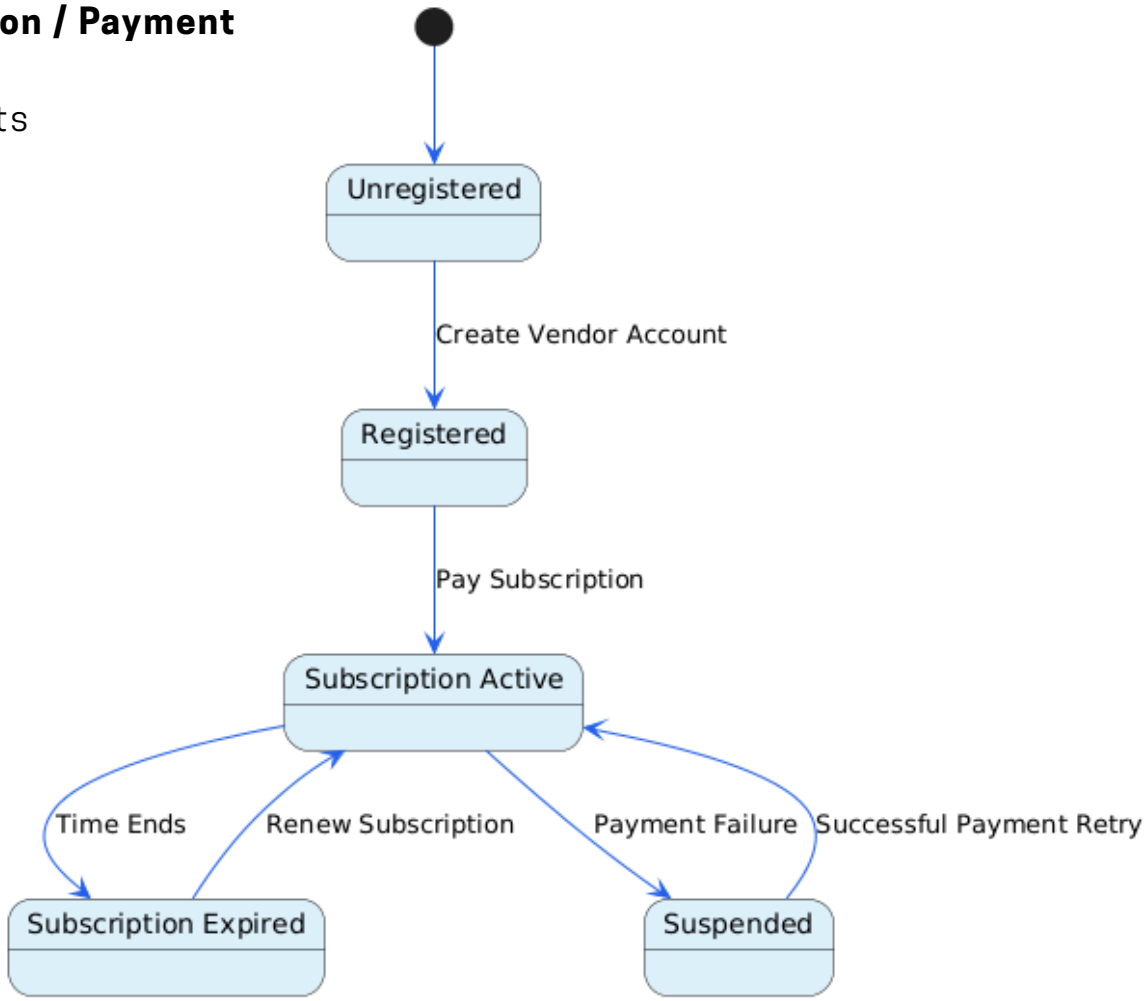
1.2 Vendor Subscription / Payment

Description

This diagram represents how vendor access changes based on registration and subscription status.

Process Flow

The process begins in Unregistered. After account creation, the vendor moves to Registered with limited access. After payment, the vendor enters Subscription Active and gains full access.



Possible transitions:

- Subscription Active → Subscription Expired (period ends)
- Subscription Expired → Subscription Active (renewal)
- Subscription Active → Suspended (payment failure)
- Suspended → Subscription Active (successful retry)

Key Observations

- Access is controlled by subscription status
- Handles normal and exception scenarios (payment failure)
- Process is cyclical, allowing renewals

Design Relevance

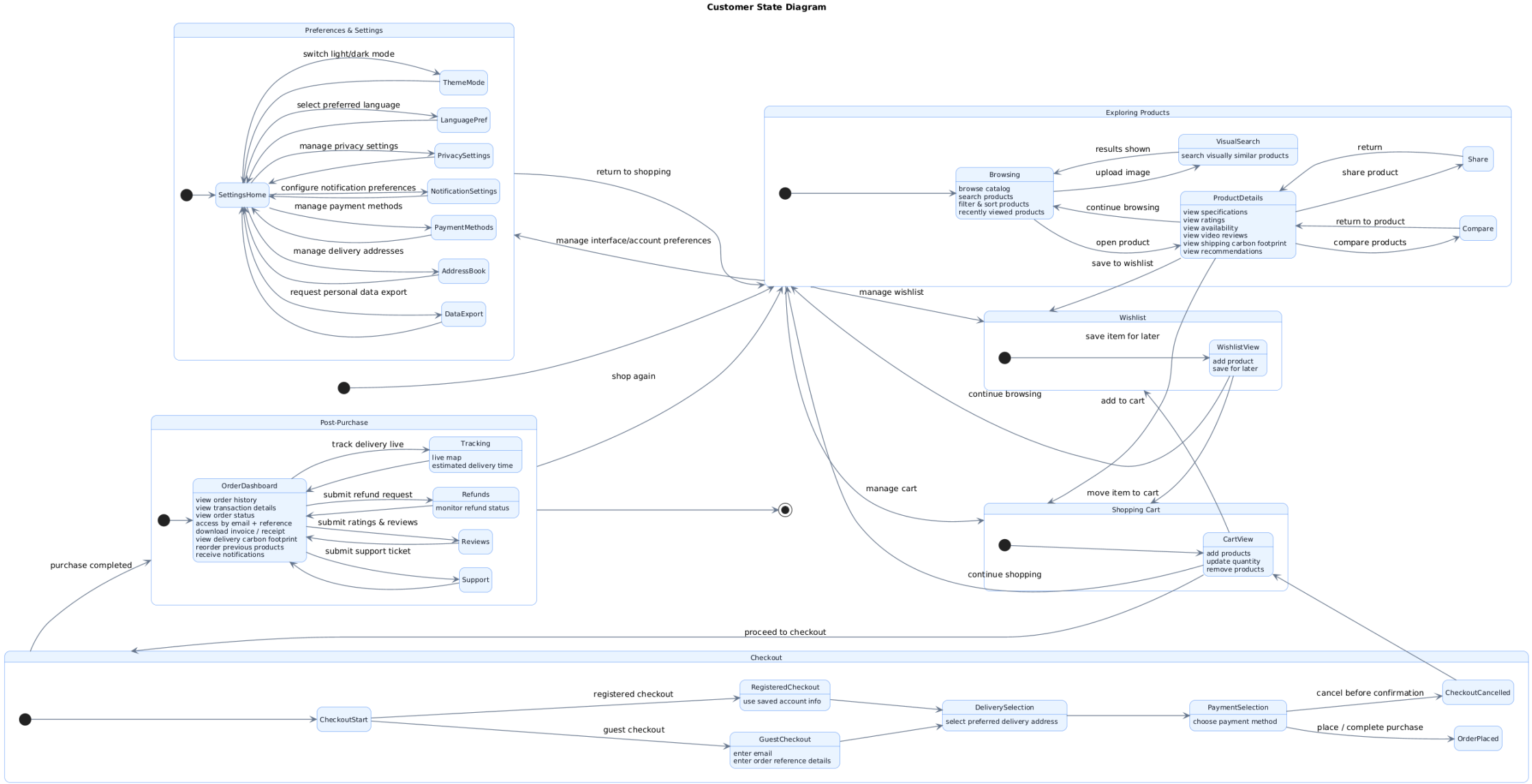
- Separates access control from operational workflows
- Supports subscription-based functionality
- Ensures vendors require valid payment status

1.3 Overall Vendor Behavior

The vendor is modeled through two independent processes:

- Order handling (operational workflow)
- Subscription management (access and payment logic)

This separation improves clarity and ensures independent handling of each aspect.


Customer State Diagram
Preferences & Settings
switch light/dark mode
ThemeMode
select preferred language
LanguagePref
manage privacy settings
PrivacySettings
SettingsHome
configure notification preferences
NotificationSettings
manage payment methods
PaymentMethods
manage delivery addresses
AddressBook
request personal data export
DataExport
return to shopping
manage interface/account preferences
Exploring Products
VisualSearch
search visually similar products
results shown
upload image
Browsing
browse catalog
search products
filter & sort products
recently viewed products
continue browsing
open product
ProductDetails
view specifications
view ratings
view availability
view video reviews
view shipping carbon footprint
view recommendations
save to wishlist
return
Share
share product
return to product
Compare
compare products
manage wishlist
Wishlist
save item for later
WishlistView
add product
save for later
continue browsing
add to cart
shop again
Post-Purchase
track delivery live
Tracking
live map
estimated delivery time
OrderDashboard
view order history
view transaction details
view order status
access by email + reference
download invoice / receipt
view delivery carbon footprint
reorder previous products
receive notifications
submit refund request
Refunds
monitor refund status
submit ratings & reviews
Reviews
submit support ticket
Support
manage cart
move item to cart
Shopping Cart
CartView
add products
update quantity
remove products
continue shopping
purchase completed
proceed to checkout
Checkout
CheckoutStart
registered checkout
RegisteredCheckout
use saved account info
guest checkout
GuestCheckout
enter email
enter order reference details
DeliverySelection
select preferred delivery address
PaymentSelection
choose payment method
cancel before confirmation
CheckoutCancelled
place / complete purchase
OrderPlaced

2. Customer Order and Shopping Flow

Description

This diagram models the complete customer journey in an e-commerce system, starting from product exploration and continuing through cart management, checkout, and post-purchase activities. It captures user interactions across browsing, personalization, purchasing, and order management.

Process Flow

The process begins in the Shopping & Discovery state, where the customer enters the system and transitions to Browsing.

Within the browsing phase, the customer can:

- Search, filter, and sort products
- Browse categories and view recently viewed products
- Upload an image to search for visually similar products
- Select a product and move to View Product

From View Product, the user can:

- View product specifications, ratings, and availability (in stock / out of stock)
- Watch video-based product reviews
- View personalized product recommendations
- View the carbon footprint of the selected shipping method
- Compare products
- Share product listings via link or social media
- Add items to the Cart
- Add or manage items in the Wishlist
- Return to browsing

At any point, the user may:

- Open the Cart, where they can update or remove items, move items to wishlist, continue shopping, or proceed to checkout
- Access the Wishlist, where they can save products for future purchase or move them to the cart
- Access Settings, where they can manage:
  - Theme (light/dark mode)
  - Language
  - Privacy settings
  - Notification preferences (order updates, delivery status, cancellations, price drops, offers, restocking)
  - Payment methods
  - Delivery addresses
  - Personal data export

*When proceeding from the cart, the system transitions to **Checkout**, where:*

- The user chooses between Guest Checkout or Logged-In Checkout
- Provides or selects a delivery address
- Selects a payment method

*The process then moves to **Payment**:*

- If successful → transitions to order completion
- If cancelled → returns to the cart

*After a successful payment, the order enters the **Post-Purchase Flow**, where the customer can:*

- View order history, order status, and transaction details
- Track delivery in real time with live map updates and estimated delivery time
- Download invoices or receipts
- Access order tracking and invoices using reference number and email (guest access)
- View the carbon footprint of completed deliveries
- Reorder previously purchased products
- Receive notifications for order updates, delivery status changes, cancellations, price drops, promotional offers, and restocking

Additional post-purchase actions include:

- Cancel → Cancelled → Refunded (including automatic refunds and refund status tracking)
- Submit reviews and ratings for purchased products
- Submit support tickets for issues related to orders, payments, or products
- The process may loop back to shopping, allowing continuous interaction with the system.

Key Observations

- The system supports both guest and authenticated users, increasing flexibility
- Multiple loops exist (browsing ↔ cart ↔ wishlist ↔ checkout), reflecting real user behavior
- The checkout process includes a decision point (guest vs logged-in)
- Payment includes cancellation handling and return to cart
- Post-purchase functionality is fully developed, including tracking, refunds, and support
- Advanced features such as visual search, recommendations, and carbon footprint tracking enhance user experience
- Users can re-enter the purchase cycle via reorder, ensuring continuity

Design Relevance

This model directly supports:

- Definition of customer-side states (browsing, cart, checkout, payment, order status, post-purchase services)
- Clear separation between browsing, purchasing, and post-purchase flows
- Integration with:
  - Authentication systems (guest vs logged-in users)
  - Payment gateways
  - Order tracking and logistics systems
  - Notification and recommendation systems

It ensures a scalable, complete, and user-centered design that covers the full lifecycle of customer interaction within the platform.

3.Guest Checkout Flow

Description

Models the guest user journey from browsing products to completing an order, capturing key decisions such as validation and payment handling.

Process Flow

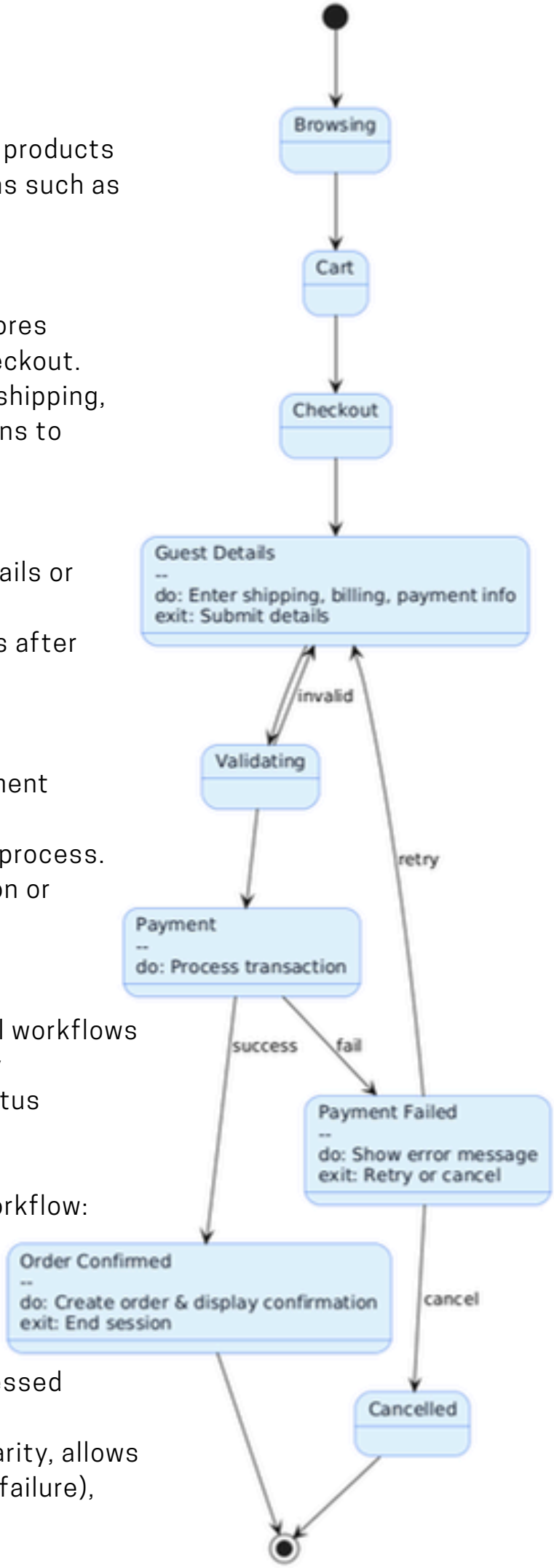
1. Browsing → Cart → Checkout – guest explores products, adds items, and proceeds to checkout.
2. GuestDetails → Validating – user submits shipping, billing, and payment info; invalid data returns to GuestDetails.
3. Payment – transaction is processed:
4. Success → OrderConfirmed
5. Fail → PaymentFailed → Retry → GuestDetails or Cancel → Cancelled
6. OrderConfirmed / Cancelled – session ends after completion or cancellation.

Key Observations

- Decision points handle validation and payment outcomes.
- Loops allow retries without restarting the process.
- Lifecycle ends only after order confirmation or cancellation

Design Relevance

- Separates access control from operational workflows
- Supports subscription-based functionality
- Ensures vendors require valid payment status

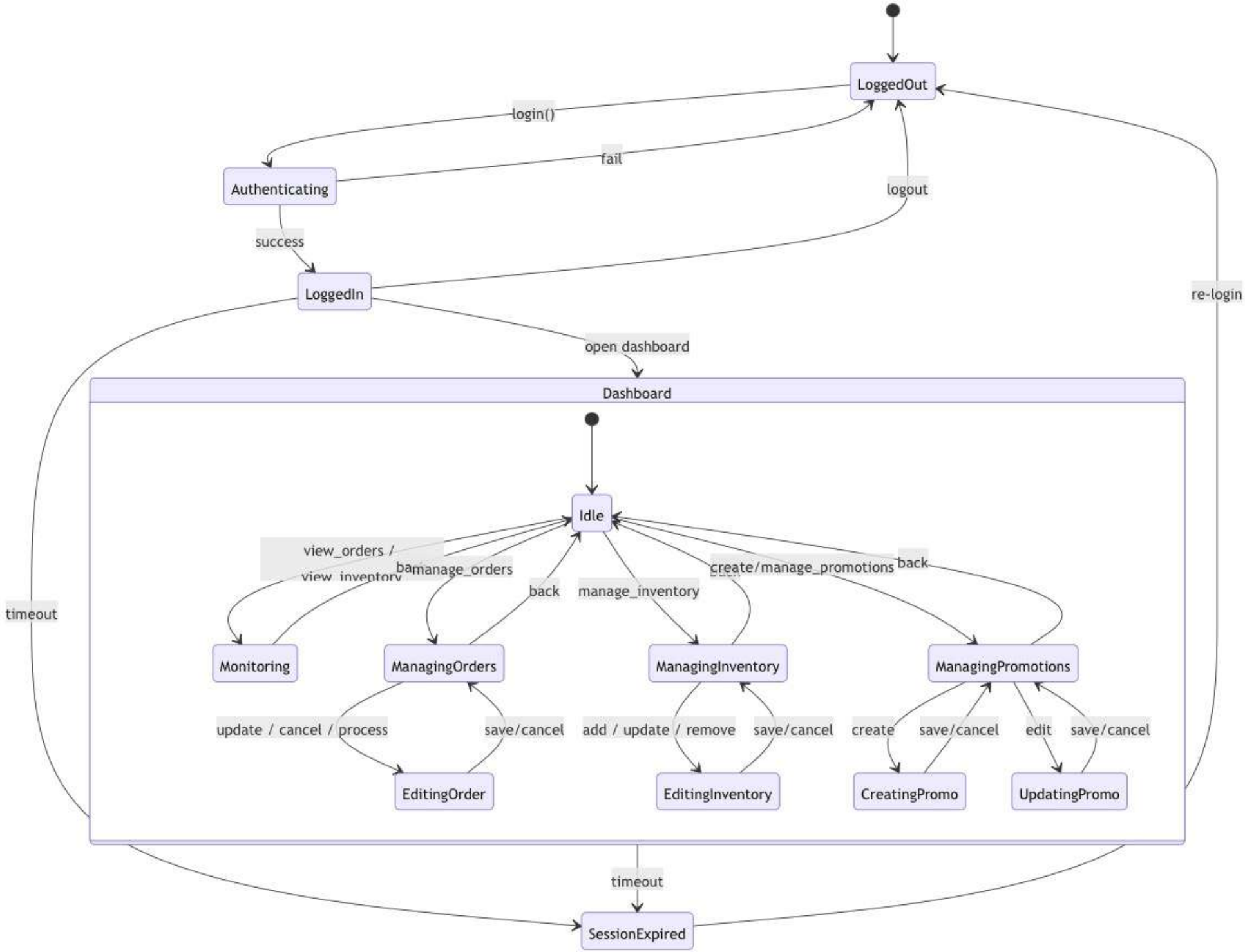


3.2 Overall Vendor Behavior

The guest user is modeled through a single workflow:

- Browsing – explore and select products.
- Cart Management – add, remove, or update items.
- Checkout and Guest Details Submission
- Payment Processing – transaction is processed
- Order Completion or Cancellation

This linear yet branching structure ensures clarity, allows handling of exceptions (invalid data, payment failure), and maintains a simple guest checkout flow.


LoggedOut
login()
fail
Authenticating
logout
success
LoggedIn
re-login
open dashboard
Dashboard
Idle
view_orders /
view_inventory
manage_orders
back
manage_inventory
create/manage_promotions
back
timeout
Monitoring
ManagingOrders
ManagingInventory
ManagingPromotions
update / cancel / process
save/cancel
add / update / remove
save/cancel
create
save/cancel
edit
save/cancel
EditingOrder
EditingInventory
CreatingPromo
UpdatingPromo
timeout
SessionExpired

4.Admin State Diagram

Description

This diagram models the behavior of the administrator within the system. It captures how the administrator performs operational tasks, navigates the dashboard, and manages different system functionalities efficiently.

Process Flow

1. The process begins in the Idle state.
2. The administrator first logs in through an authentication process and then accesses the Dashboard, which serves as the main working area.
3. From the dashboard, the administrator can transition to different tasks:
4. Monitor Orders and Inventory – viewing current system status.
5. Manage Orders – updating, approving, or canceling orders.
6. Manage Inventory – adding, editing, or removing items.
7. Handle Promotional Activities – creating or modifying promotions.
8. After completing any task, the system transitions the administrator back to the Idle state, allowing them to switch efficiently between tasks or log out.

Key Observations

- Each task is represented as a separate state.
- Transitions capture actions performed by the administrator.
- Structured flow ensures that the administrator can return to Idle before starting a new task.
- No external system interaction is required within this flow (all operations are internal).

Design Relevance

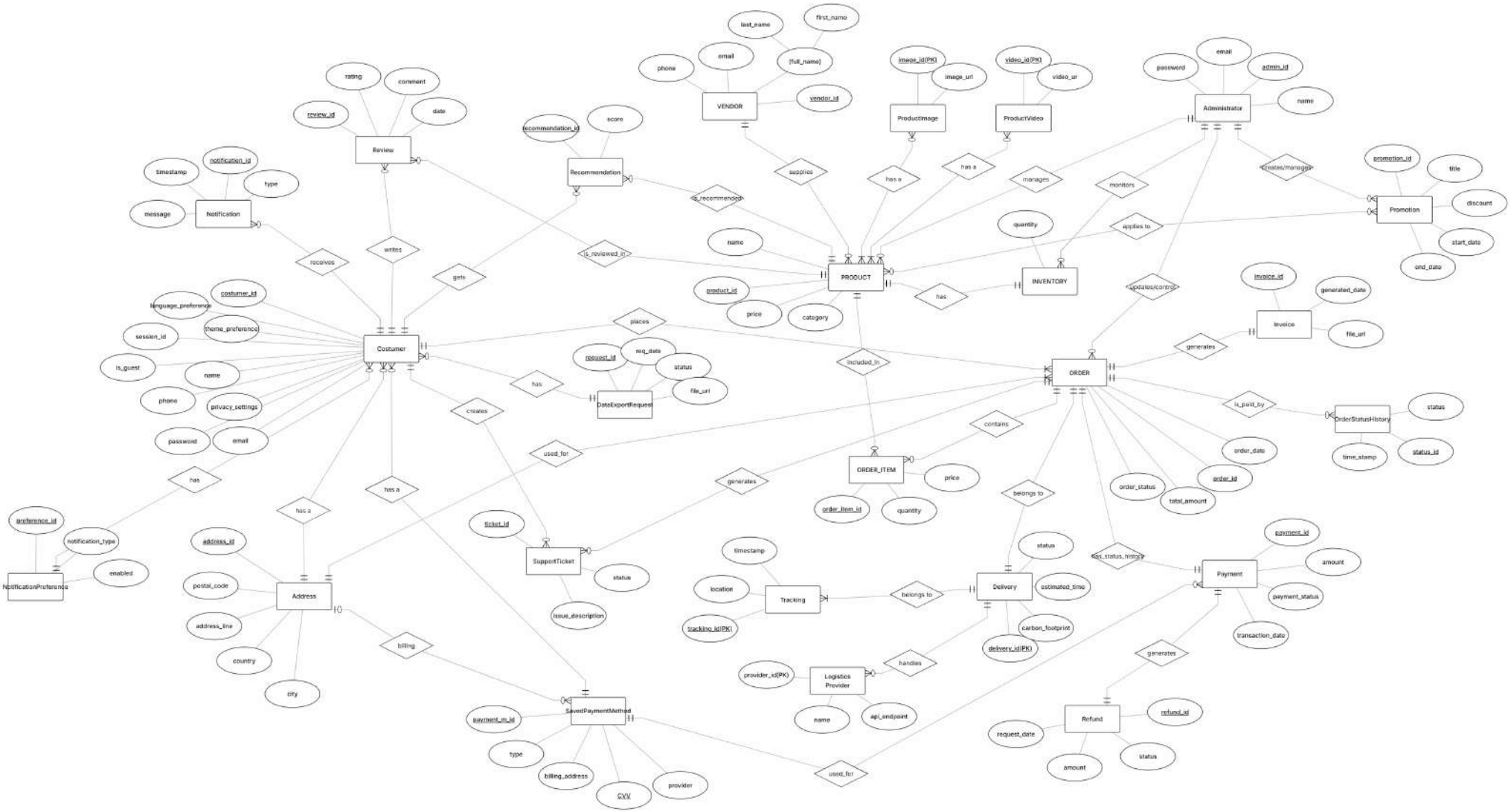
- Defines clear administrator responsibilities.
- Supports efficient navigation between tasks.
- Ensures controlled access and workflow through authentication and structured transitions.

3.2 Overall Administrator Behavior

The administrator is modeled through a structured workflow:

- Login and Authentication – access the system securely.
- Dashboard Navigation – main working area to select tasks.
- Task Execution – monitor orders/inventory, manage orders, manage inventory, handle promotions.
- Return to Idle or Logout – after each task, the administrator can switch tasks or exit the system.

This structured yet flexible flow ensures efficient task handling, allows switching between operations, and maintains system control.


Review
rating
comment
review_id
date
Recommendation
recommendation_id
score
VENDOR
phone
email
last_name
first_name
(full_name)
vendor_id
ProductImage
image_id(PK)
image_url
ProductVideo
video_id(PK)
video_ur
Administrator
password
email
admin_id
name
Notification
timestamp
notification_id
type
message
Promotion
promotion_id
title
discount
start_date
end_date
receives
writes
gets
is_recommended
is_reviewed_in
supplies
has a
has a
manages
monitors
creates/manages
applies to
PRODUCT
name
product_id
price
category
has
INVENTORY
quantity
updates/control
Costumer
costumer_id
language_preference
theme_preference
session_id
is_guest
name
phone
privacy_settings
password
email
places
has
DataExportRequest
request_id
req_date
status
file_url
creates
included_in
ORDER
generates
Invoice
invoice_id
generated_date
file_url
is_paid_by
OrderStatusHistory
status
status_id
time_stamp
order_date
order_id
total_amount
order_status
contains
ORDER_ITEM
price
order_item_id
quantity
generates
used_for
belongs to
has
has a
has a
NotificationPreference
preference_id
notification_type
enabled
Address
address_id
postal_code
address_line
country
city
billing
SupportTicket
ticket_id
status
issue_description
Tracking
timestamp
location
tracking_id(PK)
belongs to
Delivery
status
estimated_time
carbon_footprint
delivery_id(PK)
has_status_history
Payment
payment_id
amount
payment_status
transaction_date
handles
Logistics Provider
provider_id(PK)
name
api_endpoint
generates
Refund
refund_id
request_date
amount
status
SavedPaymentMethod
payment_m_id
type
billing_address
CVV
provider
used_for

ERD DIAGRAM

The ERD diagram in this project is designed to represent the overall system structure,illustrating how all entities are organized and how they interact through defined relationships. It provides a comprehensive view of the data model, showing how different components of the system are connected and how information flows between them.

User & Account Management (Elda Toska & Sindi Terihati)

Entities

Customer
- customer_id (PK)
- name
- email (unique)
- password
- phone
- language_preference
- theme_preference
- privacy_settings

Guest
- guest_id (PK)
- email
- session_id

Address
- address_id (PK)
- customer_id (FK)
- address_line
- city
- country
- postal_code

SavedPaymentMethod
- payment_method_id (PK)
- customer_id (FK)
- type (card, wallet)
- provider
- last4_digits

NotificationPreference
- preference_id (PK)
- customer_id (FK)
- type (order_update, price_drop, etc.)
- enabled (boolean)

DataExportRequest
- request_id (PK)
- customer_id (FK)
- request_date
- status

Relationships
- Customer 1—M Address
- Customer 1—M SavedPaymentMethod
- Customer 1—M NotificationPreference
- Customer 1—M DataExportRequest

Product & Catalog (Elisa Spaho)

Entities

Product
- product_id (PK)
- seller_id (FK)
- category_id (FK)
- name
- description
- price
- availability_status

Category
- category_id (PK)
- name
- parent_category_id (FK)

ProductImage
- image_id (PK)
- product_id (FK)
- image_url

ProductVideo
- video_id (PK)
- product_id (FK)
- video_ur

Bundle
- bundle_id (PK)
- seller_id (FK)
- bundle_price

BundleItem
- bundle_id (FK)
- product_id (FK)

Relationships
- Seller 1—M Product
- Category 1—M Product
- Product 1—M ProductImage
- Product 1—M ProductVideo
- Bundle M—N Product (via BundleItem)

User & Account Management (Arjola Metaliaj)

Entities

Seller
- seller_id (PK)
- name
- email
- subscription_status

PriceHistory
- address_id (PK)
- customer_id (FK)
- address_line
- city
- country
- postal_code

Inventory
- guest_id (PK)
- email
- session_id

Relationships
- Seller 1—M Product
- Product 1—1 Inventory
- Product 1—M PriceHistory

Order & Payment (Asja Xhelezi)

Entities

Order
- order_id (PK)
- customer_id (FK, nullable)
- guest_id (FK, nullable)
- address_id (FK)
- order_date
- total_amount
- status

OrderItem
- order_item_id (PK)
- order_id (FK)
- product_id (FK)
- quantity
- price
-

OrderStatusHistory
- status_id (PK)
- order_id (FK)
- status
- timestamp

Payment
- payment_id (PK)
- order_id (FK)
- payment_method_id (FK, nullable)
- amount
- payment_status
- transaction_date

Refund
- refund_id (PK)
- payment_id (FK)
- amount
- status
- request_date

Invoice
- invoice_id (PK)
- order_id (FK)
- generated_date
- file_url

Relationships
- Order 1—M OrderItem
- Product 1—M OrderItem
- Order 1—M OrderStatusHistory
- Order 1—1 Payment
- Payment 1—1 Refund
- Order 1—1 Invoice

Delivery & Logistics (Elisa Spaho)

Entities

Delivery
- product_id (PK)
- seller_id (FK)
- category_id (FK)
- name
- description
- price
- availability_status

Tracking
- tracking_id (PK)
- delivery_id (FK)
- location
- timestamp

LogisticsProvider
- provider_id (PK)
- name
- api_endpoint

Relationships
- Order 1—1 Delivery
- Delivery 1—M Tracking
- LogisticsProvider 1—M Delivery

Admin (Stilianos Troka)

Entities

Delivery
- product_id (PK)
- seller_id (FK)
- category_id (FK)
- name
- description
- price
- availability_status

Tracking
- tracking_id (PK)
- delivery_id (FK)
- location
- timestamp

LogisticsProvider
- provider_id (PK)
- name
- api_endpoint

Relationships
- Order 1—1 Delivery
- Delivery 1—M Tracking
- LogisticsProvider 1—M Delivery

Engagement & System (Ederta Kariqi)

Entities

Review
- review_id (PK)
- customer_id (FK)
- product_id (FK)
- rating
- comment
- date

Recommendation
- recommendation_id (PK)
- customer_id (FK)
- product_id (FK)
- score

Notification
- notification_id (PK)
- customer_id (FK)
- type
- message
- timestamp

SupportTicket
- ticket_id (PK)
- customer_id (FK)
- order_id (FK)
- issue_description
- status

Promotion
- promotion_id (PK)
- admin_id (FK)
- code
- discount_value
- expiry_date

Admin
- admin_id (PK)
- name
- email

AuditLog
- log_id (PK)
- user_id
- action
- timestamp

ErrorLog
- error_id (PK)
- service_name
- error_code
- timestamp
-

Relationships
- Customer 1—M Review
- Product 1—M Review
- Customer 1—M Notification
- Customer 1—M Recommendation
- Customer 1—M SupportTicket
- Order 1—M SupportTicket
- Admin 1—M Promotion

SYSTEM-WIDE

These connect all modules together:

Customer / Guest → Cart → Order
Order → Payment → Refund
Order → Delivery → Tracking
Product → CartItem / OrderItem / Review / Wishlist
Seller → Product → Inventory
Admin → Promotion
Customer → Notification / Support / Recommendation